Preface to OCP Java SE 17 Developer (Exam 1Z0-829)

Java has been around for nearly 30 years. During this time, the Java ecosystem has evolved to become the platform of choice for developing software systems, and the Java certification has evolved with it.

Part II: **OCP Java SE 17 Developer** of this guide provides extensive coverage for the following Java certifications:

- *Oracle Certified Professional (OCP), Java SE 17 Developer*, and its required exam, *Java SE 17 Developer (1Z0-829)* (for details, see **Appendix B**)
- *Oracle Certified Professional (OCP), Java SE 11 Developer,* and its required exam, *Java SE 11 Developer (1Z0-819)* (for details, see **Appendix C**)

The exam syllabus is defined by a set of *exam objectives*. The exam objectives have evolved to become more high-level, with a smattering of specific topics, leaving plenty of room for interpretation as to which topics are covered by an exam objective. The scope of the exam objectives is thus very difficult to define, and entails exhaustive coverage of the Java language features and its many APIs to be on the safe side. This factor is the major cause of the considerable size of **Part II** in this guide.

Our approach has not changed from the one we employed for our previous books, mainly because it has proved successful. The emphasis remains on analyzing code scenarios, rather than esoteric syntax of individual language constructs. The exam continues to require actual experience with the language, not just mere recitation of facts. We still claim that proficiency in the language is the key to success.

Since the exam emphasizes the core features of Java, we provide provides in-depth coverage of topics related to those features. **Part II** is also no different from our previous books in one other important aspect: It is a one-stop resource, providing a mixture of theory and practice that enables readers to prepare for the exam. It can be used to learn Java and to prepare for the exam. After the exam is passed, it can also be used as a handy reference for the Java language.

All elements found in our previous books (e.g., examples, figures, tables, review questions, mock exam questions) can be found in this one as well. We continue to use UML

(Unified Modeling Language) extensively to illustrate concepts and language constructs, and all numbered examples continue to be complete Java programs ready for experimentation.

About Part II: OCP Java SE 17 Developer

Part II provides extensive coverage of the core features of the Java programming language and its core application programming interface (API). It is primarily intended for professionals who want to prepare for the *Java SE 17* or the *Java SE 11 Developer Exam*, but it is readily accessible to any programmer who wants to master the language. For both purposes, it provides in-depth coverage of essential features of the language and its core API.

The demand for well-trained and highly skilled Java programmers remains unabated. Oracle offers many Java certifications that professionals can take to validate their skills (see **https://education.oracle.com**). The certification provides members of the IT industry with a standard to use when hiring such professionals, and it allows professionals to turn their Java skills into credentials that are important for career advancement.

Part II provides extensive coverage of all the objectives defined by Oracle for both the *Java SE 17* and *Java SE 11 Developer Exams*. The exam objectives are selective, however, and do not include many of the essential features of Java. **Part II** covers many additional topics that every Java programmer should master to be truly proficient. In this regard, **Part II** is a comprehensive primer for learning the Java programming language. After mastering the language by working through **Part II**, the reader can confidently sit for the exams.

Part II is *not* a complete reference for Java, as it does not attempt to list every member of every class from the Java SE Platform API documentation. The purpose is not to document the Java SE Platform API. The emphasis is more on the Java programming language features—their syntax and correct usage through code examples—and less on teaching programming techniques.

Part II assumes little background in programming. We believe the exam is accessible to any programmer who works through **Part II**. A Java programmer can easily skip over material that is well understood and concentrate on parts that need reinforcing, whereas a programmer new to Java will find the concepts explained from basic principles.

Each topic is explained and discussed thoroughly with examples, and is backed by review questions to reinforce the concepts. This guide is not biased toward any particular platform, but provides platform-specific details where necessary.

Using Part II: OCP Java SE 17 Developer

The reader can choose a linear or a nonlinear route through **Part II**, depending on their programming background. Non-Java programmers wishing to migrate to Java can read **Chapter 1**, which provides a short introduction to object-oriented programming concepts, and the procedure for compiling and running Java applications. For those preparing for any of the Developer exams, **Part II** has a separate appendix (**Appendix A**) providing all the pertinent information on preparing for and taking the exams.

Cross-references are provided where necessary to indicate the relationships among the various constructs of the language. To understand a language construct, all pertinent details are provided where the construct is covered, but in addition, cross-references are provided to indicate its relationship to other constructs. Sometimes it is necessary to postpone discussion of certain aspects of a topic if they depend on concepts that have not yet been introduced. A typical example is the consequences of object-oriented programming concepts (e.g., inheritance) on the member declarations that can occur in a class. This approach can result in forward references in the initial chapters of **Part II**.

The table of contents; listings of tables, examples, and figures; and a comprehensive index facilitate locating topics discussed in **Part II**.

In particular, we draw attention to the following features in **Part II**:



Chapter Topics

Each chapter starts with a short summary of the topics covered in the chapter, pointing out the major concepts that are introduced.

Developer Exam Objectives	
[0.1]	Exam objectives that are covered in each chapter are stated clearly at the beginning of every chapter.
[0.2]	The number in front of the objective identifies the exam objective, as defined by Oracle. The objectives are organized into major sections, detailing the syllabus for the exam.
[0.3]	The objectives for the *Java SE 17* and *Java SE 11 Developer Exams* are reproduced verbatim in **Appendix B** and **Appendix C**, respectively. These ap-

pendices also map each exam objective to relevant chapters and sections in **Part II**.

Supplementary Topics

- Supplementary topics are Java topics that are *not* on the exam per se, but which the candidate is expected to know.
- Any supplementary topic is listed as a bullet at the beginning of the chapter.



Review Questions

Review questions are provided after every major topic to test and reinforce the material. The review questions predominantly reflect the kind of multiple-choice questions that can be asked on the actual exam. On the exam, the exact number of answers to choose for each question is explicitly stated. The review questions in **Part II** follow that practice.

Many questions on the actual exam contain code snippets with line numbers to indicate that complete implementation is not provided, and that the necessary missing code to compile and run the code snippets can be assumed. The review questions in **Part II** provide complete code implementations where possible, so that the code can be readily compiled and run.

Annotated answers to the review questions are provided in **Appendix D**.

Example 0.1 *Example Source Code*

We encourage readers to experiment with the code examples to reinforce the material covered in **Part II**. The source code for the examples can be downloaded from the companion website of **Part II** (see **p. xli**), and readily imported into Eclipse IDE.

Java code in the guide is presented in `a monospaced font`. Lines of code in the examples or in code snippets are referenced in the text by a number, which is specified by using a single-line comment in the code. For example, in the following code snippet, the call to the method `doSomethingInteresting()` at (1) does something interesting:

Click here to view code image

```
// ...
doSomethingInteresting();              // (1)
// ...
```

Names of classes and interfaces start with an uppercase letter. Names of packages, variables, and methods start with a lowercase letter. Constants are in all uppercase letters. Interface names begin with the prefix `I`, when it makes sense to distinguish them from class names. Coding conventions are followed, except when we have had to deviate from these conventions in the interest of space or clarity.

Mock Exam

The mock exam for Java SE 17 in **Appendix E** should be attempted when the reader feels confident about the topics on the exam. It is highly recommended to read **Appendix A** before attempting the mock exam, as **Appendix A** contains pertinent information about the questions to expect on the actual exam. Each multiple-choice question in the mock exam explicitly states how many answers are applicable for a given question, as is the case on the actual exam. Annotated answers to the questions in the mock exam are provided in **Appendix F**.

Java SE Platform API Documentation

A vertical gray bar is used to highlight methods and fields found in the classes of the Java SE Platform API.

Any explanation following the API information is also similarly highlighted.

To obtain the maximum benefit from using **Part II** in preparing for the *Java SE 17* and *Java SE 11 Developer Exams*, we strongly recommend installing the latest version (Release 17 or newer) of the JDK and its accompanying API documentation. **Part II** focuses solely on Java SE 17 and Java SE 11, and does not acknowledge other versions.

Website for Part II: OCP Java SE 17 Developer

Part II is backed by a website:

Click here to view code image

https://www.mughal.no/jse17ocp/

Auxiliary material on the website includes the following:

- Source code for all the examples in **Part II**
- Annotated answers to the reviews questions in **Part II**
- Annotated answers to the mock exam in **Part II**
- Table of contents, sample chapters, and index from **Part II**
- Content specific for the *Java SE 17* and *Java SE 11 Developer Exams*
- Errata for **Part II**
- Links to miscellaneous Java resources (e.g., certification, discussion groups, and tools)

Information about the Java Standard Edition (SE) and its documentation can be found at the following website:

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www.oracle.com/technetwork/java/javase/overview/index.html

The current authoritative technical reference for the Java programming language, *The Java® Language Specification: Java SE 17 Edition*, can be found at this website:

Click here to view code image

http://docs.oracle.com/javase/specs/index.html

Acknowledgments

First of all, we would like to acknowledge the contribution that Rolf W. Rasmussen at vizrt made in earlier Java certification books that we wrote together. Some of that contribution still permeates this edition as well.

At Pearson, senior executive editor Greg Doench was once again at the helm. He managed the long process of publishing this guide. Senior content producer Julie Nahil was again the in-house contact at Pearson, professionally and efficiently managing production of **Part II**. Audrey Doyle did a truly marvelous job copyediting **Part II**. She scrutinized every sentence with a meticulous eye—and taught us a thing or two about the

English language. The folks at codeMantra did the critical proofreading, endorsing **Part II** with a quality assurance stamp. Rob Mauhar at The CIP Group performed the typesetting wizardry necessary to give **Part II** its professional look. Our sincere thanks to Greg, Julie, Audrey, Rob, the folks at codeMantra, and all those behind the scenes at Pearson, who helped to make this publication see the light of day.

For the technical review of **Part II**, we were lucky to have two Java gurus who agreed to take on the task.

- Mikalai Zaikin is a lead Java developer at IBA Lithuania, and is currently located in Vilnius. He has helped Oracle with development of Java certification exams and has also been a technical reviewer of several Java certification books. He also contributes to the Java Quiz column for Oracle's Java Magazine in collaboration with Simon Roberts.
- Ankit Garg is currently Vice President at Morgan-Stanley, Bengaluru, India, and a Sheriff (i.e., Moderator) for various Java Forums at **coderanch.com** . He is a Java developer with many years' experience in several Java-related technologies (e.g., Java SE, Java EE, and Spring). He is also a certified Java Programmer and Web Component Developer.

Without doubt, both Mikalai and Ankit have an eye for detail. It is no exaggeration to say that their feedback has been invaluable in improving the quality of **Part II** at all levels. Our most sincere thanks to both Mikalai and Ankit for the many excellent comments and suggestions, and above all, for weeding out numerous pesky errors in the manuscript.

Kristian Berg and Morten Nygaard Åsnes were kind enough to review two chapters each and provide useful feedback, for which we are very grateful.

Great effort has been made to eliminate mistakes and errors in **Part II**. We accept full responsibility for any remaining oversights. We hope that when our diligent readers find any, they will bring them to our attention.

Without family support this edition would still be a fantasy, and we are very grateful to our families for putting up with us when we were burning the midnight oil. Hopefully we can get our family life back now that this publication is out the door.

With that, dear reader, we wish you all the best should you decide to go down the path of Java certification. May your loops terminate and your exceptions get caught!

—Khalid A. Mughal
November 5, 2024

Bergen, Norway

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